

Minimal Unitary Realizations Can Agree Locally and Disagree on a Disconnected Component

Counterexample packet for arXiv:2509.08921

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Status: candidate full counterexample, likely valid. There are two minimal, analytically equivalent univariate realizations whose operators are unitary with spectrum $\mathbb{T}$, but whose transfer functions differ by exactly $-1/z$ on $\mathbb{C} \setminus \overline{\mathbb{D}}$. This supplies the example requested in Remark 2.4 of the source paper.

1 Source and exact question

The source is A. Karoobi, R. T. W. Martin, and M. Tornes, *Operator realizations about a matrix-centre*, arXiv:2509.08921 [1]. For a bounded operator A on a Hilbert space and vectors b, c , the scalar transfer function is

$$f_A(z) = b^*(I - zA)^{-1}c.$$

The realization is *minimal* when c is A -cyclic and b is A^* -cyclic. Two realizations are analytically equivalent at 0 exactly when their moments $b^*A^n c$ agree for every $n \geq 0$; equivalently, their matrix-level transfer power series agree near 0.

Remark 2.4 on source PDF page 8 asks whether there exist two minimal, analytically equivalent realizations with disconnected invertibility domains which define different analytic functions on a component disconnected from 0.

2 Proof intuition

The source's bilateral-shift example fails minimality because a point mass at one Fourier mode sees only half of the bilateral orbit. We replace that point mass by Lebesgue measure restricted to a proper measurable part of the unit circle. On any proper piece whose complement has positive measure, Hardy boundary uniqueness makes both analytic and anti-analytic monomials dense; this gives the two cyclicity conditions and hence minimality.

Split the circle into two pieces E and F , each occurring in every arc. The first realization uses $L^2(F)$ and output vector $\bar{\zeta}$, while the second uses $L^2(E)$ and output vector $-\bar{\zeta}$. Their nonnegative moments add up to the positive Fourier coefficients of Lebesgue measure, hence cancel. On the exterior component the missing Laurent coefficient is the constant Fourier coefficient instead, and it survives as $-1/z$.

3 Two elementary lemmas

Let m denote normalized Lebesgue measure on $\mathbb{T}$.

Lemma 1 (Everywhere-thick measurable partition). *There is a measurable partition $\mathbb{T} = E \dot{\cup} F$ such that*

$$m(E \cap I) > 0 \quad \text{and} \quad m(F \cap I) > 0$$

equivalent at Y , written $(A, b, c)_Y \sim (A', b', c')_Y$ or $(A, b, c) \sim_Y (A', b', c')$, if they define the same quantized transfer function on a uniformly open neighbourhood of Y .

In general, even if $f \sim (A, b, c)_Y$, $g \sim (A', b', c')_Y$ and $(A, b, c)_Y \sim (A', b', c')_Y$, it may be that $f(X) \neq g(X)$ for some $X \in \mathcal{D}^Y(A) \cap \mathcal{D}^Y(A')$, as the following example (with $Y = 0$, $d = 1$) shows.

Example 2.3. Let U be the bilateral shift on $\ell^2(\mathbb{Z})$, and consider the realization (U, e_0, e_0) . Then, $\sigma(U) = \partial\mathbb{D}$, so that the resolvent set of U , $\sigma(U)^c = \mathbb{C} \setminus \sigma(U)$, is disconnected. Hence $\mathcal{D}_1(U) = \mathbb{C} \setminus \partial\mathbb{D}$ is also disconnected. Then, for $|z| < 1$,

$$e_0^*(I - zU)^{-1}e_0 = \sum_{j=0}^{\infty} e_0^*U^j e_0 z^j = 1,$$

while for $|z| > 1$,

$$\begin{aligned} e_0^*(I - zU)^{-1}e_0 &= -z^{-1}e_0^*U^*(I - z^{-1}U^*)^{-1}e_0 \\ &= -\sum_{j=0}^{\infty} z^{-j-1}e_0^*U^{*(j+1)}e_0 = 0. \end{aligned}$$

That is, the realization $(U, e_0, e_0) \sim f$ defines the analytic function which is 1 in $\mathbb{D}$ and 0 in $\mathbb{C} \setminus \overline{\mathbb{D}}$. On the other hand, the one-dimensional realization $(0, 1, 1)$ defines the constant function 1 everywhere. This shows that realizations do not need to evaluate to the same analytic function on the intersection of their invertibility domains, in general.

Remark 2.4. Later, we will define a realization (A, b, c) to be *minimal*, if c is A -cyclic and b is A^* -cyclic. (A minimal realization is, in some sense as 'small' as possible.) In the above example, (U, e_0, e_0) is a non-minimal realization, but $(0, 1, 1)$ is minimal. We do not know whether one can construct an example of two minimal and analytically equivalent realizations, $(A, b, c) \sim_0 (A', b', c')$, with disconnected invertibility domains, so that they define different analytic functions on a connected component disconnected from 0.

Figure 1: Source PDF page 8: Example 2.3 and the exact question in Remark 2.4.

for every nonempty open arc $I \subset \mathbb{T}$.

Proof. Enumerate a countable basis of open arcs as $(I_k)_{k \geq 1}$. Inductively, inside I_k and away from the finitely many compact sets already selected, choose two disjoint compact nowhere-dense sets K_k, L_k of positive measure. This is possible by placing two small fat Cantor sets in an open subarc of the remaining set. Put $E = \bigcup_{k \geq 1} K_k$ and $F = \mathbb{T} \setminus E$. Every future K_j is chosen disjoint from every earlier L_k , so $L_k \subset F$. Every open arc contains some basis arc I_k and therefore contains positive-measure subsets $K_k \subset E$ and $L_k \subset F$. $\square$

Lemma 2 (One-sided polynomial density). *If $G \subset \mathbb{T}$ is measurable and $m(\mathbb{T} \setminus G) > 0$, then both*

$$\text{span}\{\zeta^n : n \geq 0\} \quad \text{and} \quad \text{span}\{\bar{\zeta}^n : n \geq 0\}$$

are dense in $L^2(G, m)$.

Proof. Suppose $h \in L^2(G)$ is orthogonal to every ζ^n , $n \geq 0$. Extend $\mathbf{1}_G \bar{h}$ by zero to $\mathbb{T}$. Its Fourier coefficients of index at most zero vanish, so it is the boundary function of an element of zH^2 . It vanishes on $\mathbb{T} \setminus G$, a set of positive measure. Hardy boundary uniqueness forces it to vanish identically; see, for example, [2, Chapter II]. Hence $h = 0$. This proves density of analytic polynomials. Complex conjugation gives the anti-analytic assertion. $\square$

4 The counterexample

Choose E, F as in Lemma 1. For $G \in \{E, F\}$ let

$$\mathcal{H}_G = L^2(G, m), \quad (U_G h)(\zeta) = \zeta h(\zeta).$$

Define two realizations by

$$(A, b, c) = (U_F, \bar{\zeta}, \mathbf{1}_F), \quad (A', b', c') = (U_E, -\bar{\zeta}, \mathbf{1}_E). \quad (1)$$

Theorem 1. *The two realizations in (1) are minimal and analytically equivalent at 0. Both scalar invertibility domains are*

$$\mathbb{C} \setminus \mathbb{T} = \mathbb{D} \dot{\cup} (\mathbb{C} \setminus \bar{\mathbb{D}}).$$

If f_F and f_E are their scalar transfer functions, then

$$f_F(z) = f_E(z) \quad (|z| < 1), \quad f_F(z) - f_E(z) = -\frac{1}{z} \quad (|z| > 1).$$

Thus they give the example requested in Remark 2.4 of [1].

Proof. Each U_G is unitary. Since every open arc has positive G -measure, the essential range of the coordinate function on G is all of $\mathbb{T}$. Thus $\sigma(U_E) = \sigma(U_F) = \mathbb{T}$, and $I - zU_G$ is invertible exactly when $|z| \neq 1$.

By Lemma 2, $\mathbf{1}_G$ is U_G -cyclic. The adjoint is $U_G^* = M_{\bar{\zeta}}$, and

$$(U_F^*)^n \bar{\zeta} = \bar{\zeta}^{n+1}, \quad (U_E^*)^n (-\bar{\zeta}) = -\bar{\zeta}^{n+1}.$$

The shifted anti-analytic monomials are dense because multiplication by $\bar{\zeta}$ is unitary and the unshifted anti-analytic monomials are dense by Lemma 2. Hence both realizations are minimal.

For every $n \geq 0$, their Markov parameters are

$$\begin{aligned} b^* A^n c &= \int_F \zeta^{n+1} dm(\zeta), \\ b'^* A'^n c' &= - \int_E \zeta^{n+1} dm(\zeta). \end{aligned}$$

Because $E \dot{\cup} F = \mathbb{T}$ modulo null sets,

$$b^* A^n c - b'^* A'^n c' = \int_{\mathbb{T}} \zeta^{n+1} dm(\zeta) = 0.$$

The scalar transfer series therefore agree on $\mathbb{D}$. More strongly, at every matrix level and for every matrix Z with $\|Z\| < 1$, both quantized transfer functions are

$$\sum_{n \geq 0} Z^n (b^* A^n c),$$

so the realizations are analytically equivalent at 0 in the precise sense of the source paper.

Finally, for any $|z| \neq 1$,

$$\begin{aligned} f_F(z) - f_E(z) &= \int_F \frac{\zeta}{1 - z\zeta} dm(\zeta) + \int_E \frac{\zeta}{1 - z\zeta} dm(\zeta) \\ &= \int_{\mathbb{T}} \frac{\zeta}{1 - z\zeta} dm(\zeta). \end{aligned}$$

For $|z| < 1$, the geometric series has only strictly positive Fourier modes, so the last integral is zero. For $|z| > 1$,

$$\frac{\zeta}{1 - z\zeta} = -\frac{1}{z} \sum_{k=0}^{\infty} z^{-k} \zeta^{-k},$$

uniformly on $\mathbb{T}$, and integration leaves only the $k = 0$ term. The difference is therefore $-1/z$, as claimed. $\square$

5 Verification, novelty, and scope

Verification. The accompanying verification report audits separately the measurable partition, the two cyclicity statements, the spectrum calculation, equality at every matrix level, and the exterior Laurent coefficient. No numerical computation is used; all identities are exact.

Bounded novelty search. The four run indexes were searched for arXiv:2509.08921, the exact title, “minimal and analytically equivalent realizations,” disconnected invertibility domains, unitary realizations, and Markov parameters. Exact-pharse and close arXiv/web searches through 27 August 2026 returned the source and general realization literature, but no later paper or preprint answering Remark 2.4. Novelty is plausible, not certified.

Scope and review. This is a full affirmative answer to the existence question in Remark 2.4: both realizations are minimal, both domains have the required disconnected component, and their global values differ there. It does not address the separate compact matrix-centre realization question in Section 9 of the source. Human review should focus on Lemma 2 and the sign convention in the two moment formulas.

References

- [1] A. Karoobi, R. T. W. Martin, and M. Tornes, *Operator realizations about a matrix-centre*, [arXiv:2509.08921](#), 2025.
- [2] P. L. Duren, *Theory of H^p Spaces*, Pure and Applied Mathematics 38, Academic Press, 1970; Dover reprint, 2000.